

10 questions

1. what is EV
2. explain regeneration in ev
3. what will be the charging time
4. if then how is ev is better then ic engine
5. what type of motor is used in ev
6. explain the infrastructure required in satisfactory operation of ev
7. where should be the charging station for long distance
8. how to determine the rating of charging station
9. what are the factor to be considered upon the installation on ev station
10. what should be the price for fast charging and normal charging

### **Activities**

#### **Activity 1: Introduction to Electric Vehicles (EV)**

**Objective:** Understand the concept and working of Electric Vehicles.

**Activity:** Research the definition of an EV, identify its major components (battery, motor, controller, charger), and explain how it differs from a conventional vehicle.

**Outcome:** Students will understand the basics of EV technology.

### **Answer:**

#### **Activity 1: Introduction to Electric Vehicles (EV)**

##### **Objective**

To understand the concept, working principle, components, advantages, and applications of Electric Vehicles (EVs), and compare them with conventional Internal Combustion Engine (ICE) vehicles.

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##### **Introduction**

An **Electric Vehicle (EV)** is a vehicle that uses one or more **electric motors** for propulsion instead of relying primarily on an internal combustion engine (ICE). The electric motor is powered by electricity stored in a rechargeable battery pack. Unlike conventional vehicles, EVs do not require petrol or diesel to operate (except in hybrid variants), making them cleaner and more energy-efficient.

With increasing concerns about air pollution, climate change, and the depletion of fossil fuels, EVs have emerged as one of the most promising alternatives for sustainable transportation. Governments around the world are encouraging the adoption of EVs through subsidies, tax benefits, and investments in charging infrastructure.

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##### **Definition of an Electric Vehicle**

An **Electric Vehicle (EV)** is a vehicle that uses electrical energy stored in batteries to drive an electric motor, which converts electrical energy into mechanical energy to propel the vehicle. Depending on the type of EV, the battery is charged using an external power source, regenerative braking, or a combination of both.

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##### **Working Principle of an Electric Vehicle**

The working of an EV can be understood through the following sequence:

##### **Step 1: Battery Charging**

The battery pack is charged using an external charging station or home charger. During charging, electrical energy from the grid is stored as chemical energy inside the battery.

#### **Step 2: Power Delivery**

When the driver presses the accelerator, the Battery Management System (BMS) and motor controller regulate the amount of electrical power supplied from the battery to the motor.

#### **Step 3: Motor Operation**

The electric motor converts electrical energy into mechanical energy (rotational motion), which drives the wheels through a reduction gear or transmission.

#### **Step 4: Vehicle Movement**

The generated torque rotates the wheels, moving the vehicle forward or backward depending on the driver's input.

#### **Step 5: Regenerative Braking**

When the driver applies the brakes or releases the accelerator, the electric motor acts as a generator. It converts part of the vehicle's kinetic energy back into electrical energy, which is stored in the battery, improving overall efficiency.

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### **Major Components of an Electric Vehicle**

#### **1. Battery Pack**

The battery pack is the primary energy source of an EV. It stores electrical energy required to operate the motor and other electrical systems.

##### **Functions**

- Stores electrical energy.
- Supplies power to the motor.
- Determines the driving range of the vehicle.

##### **Common Battery Types**

- Lithium-ion (most widely used)
- Lithium Iron Phosphate (LFP)
- Nickel Metal Hydride (NiMH)

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#### **2. Electric Motor**

The electric motor converts electrical energy into mechanical energy to rotate the wheels.

##### **Functions**

- Produces torque instantly.
- Provides smooth acceleration.
- Drives the vehicle without fuel combustion.

##### **Common Motor Types**

- Permanent Magnet Synchronous Motor (PMSM)
- Brushless DC Motor (BLDC)
- AC Induction Motor
- Switched Reluctance Motor (SRM)

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#### **3. Motor Controller (Inverter)**

The motor controller regulates the power supplied to the motor based on the driver's acceleration and braking inputs.

##### **Functions**

- Controls motor speed.

- Controls motor torque.
- Converts DC power from the battery into AC power for AC motors.
- Enables regenerative braking.

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#### 4. On-board Charger (OBC)

The on-board charger converts AC electricity from the grid into DC electricity required to charge the battery.

##### Functions

- Charges the battery safely.
- Controls charging current and voltage.
- Protects the battery during charging.

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#### 5. Battery Management System (BMS)

The BMS monitors and protects the battery pack.

##### Functions

- Monitors battery voltage and current.
- Balances the charge among battery cells.
- Prevents overcharging and deep discharge.
- Monitors battery temperature.

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#### 6. DC-DC Converter

The DC-DC converter steps down the high battery voltage (typically 300–800 V) to 12 V or 24 V for auxiliary systems.

##### Functions

- Powers headlights, infotainment, wipers, and other accessories.
- Charges the auxiliary battery.

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#### 7. Charging Port

The charging port connects the EV to an external charging station.

##### Types

- AC Charging Port
- DC Fast Charging Port

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#### 8. Transmission (Reduction Gear)

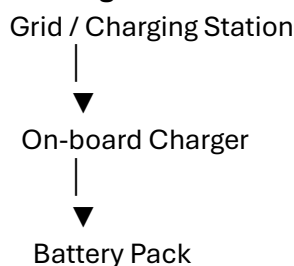
Unlike conventional vehicles with multi-speed gearboxes, most EVs use a single-speed reduction gear.

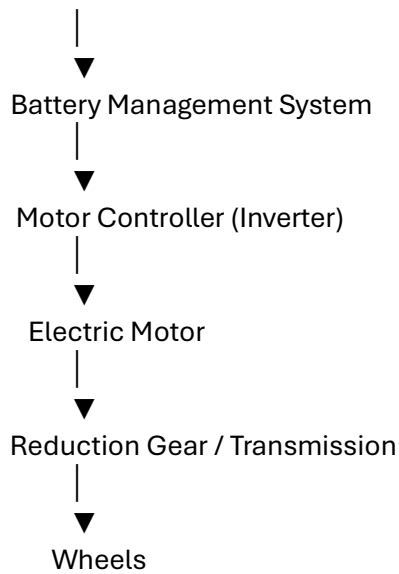
##### Functions

- Transfers motor torque to the wheels.
- Reduces motor speed while increasing torque.

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#### Block Diagram of an Electric Vehicle





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### Comparison Between EV and Conventional (ICE) Vehicle

| Feature              | Electric Vehicle (EV) | Internal Combustion Engine (ICE) Vehicle     |
|----------------------|-----------------------|----------------------------------------------|
| Energy Source        | Electricity           | Petrol/Diesel                                |
| Engine               | Electric Motor        | IC Engine                                    |
| Fuel Consumption     | No fuel               | Uses fossil fuels                            |
| Tailpipe Emissions   | Zero                  | High CO <sub>2</sub> and pollutant emissions |
| Efficiency           | 85–95%                | 25–40%                                       |
| Maintenance          | Low                   | High                                         |
| Noise                | Very low              | High                                         |
| Operating Cost       | Lower                 | Higher                                       |
| Regenerative Braking | Available             | Not available                                |
| Instant Torque       | Yes                   | No                                           |

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### Advantages of Electric Vehicles

- Zero tailpipe emissions, reducing air pollution.
- Higher energy efficiency than ICE vehicles.
- Lower running and maintenance costs.
- Quiet and smooth operation.
- Instant torque for better acceleration.
- Regenerative braking improves energy efficiency.
- Can be charged using renewable energy sources.
- Reduced dependence on fossil fuels.

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### **Disadvantages of Electric Vehicles**

- Higher initial purchase cost.
- Longer charging time compared to refueling.
- Limited charging infrastructure in some regions.
- Driving range depends on battery capacity.
- Battery performance can reduce in extreme temperatures.
- Battery replacement is expensive.

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### **Applications of Electric Vehicles**

- Personal transportation (cars and two-wheelers)
- Public transportation (electric buses)
- Delivery and logistics vehicles
- Commercial fleets
- Industrial vehicles
- Airport ground support equipment
- Ride-sharing and taxi services

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### **Outcome**

After completing this activity, students will be able to:

- Define an Electric Vehicle (EV) and explain its importance.
- Describe the working principle of an EV.
- Identify the major components of an EV and explain their functions.
- Differentiate between Electric Vehicles and Internal Combustion Engine (ICE) vehicles.
- List the advantages, disadvantages, and applications of Electric Vehicles.
- Understand why EVs are considered a key technology for sustainable transportation.

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### **Conclusion**

Electric Vehicles are transforming the transportation sector by offering a cleaner, more efficient, and environmentally friendly alternative to conventional vehicles. Their operation is based on the conversion of electrical energy stored in batteries into mechanical energy using electric motors. Components such as the battery pack, motor, motor controller, Battery Management System (BMS), charger, and DC-DC converter work together to ensure efficient vehicle performance. Although EVs face challenges such as charging time and infrastructure availability, their high efficiency, lower operating costs, and zero tailpipe emissions make them a crucial solution for achieving sustainable mobility and reducing dependence on fossil fuels.

## Activity 2: Regenerative Braking in Electric Vehicles

### Objective

To understand the concept, working principle, advantages, and importance of regenerative braking in electric vehicles (EVs).

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### Introduction

**Regenerative braking** is an energy recovery system used in electric vehicles that converts the vehicle's **kinetic energy** into **electrical energy** during braking or deceleration. Instead of wasting this energy as heat, it is stored back in the battery, improving the overall efficiency and driving range of the EV.

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### Working Principle

During normal driving, the battery supplies electrical energy to the electric motor, which rotates the wheels.

When the driver applies the brakes or releases the accelerator:

1. The electric motor stops acting as a motor and starts acting as a **generator**.
  2. The rotating wheels continue to spin the motor.
  3. The motor converts the vehicle's kinetic energy into electrical energy.
  4. The generated electricity is sent back to the battery for storage.
  5. The vehicle slows down while simultaneously recovering energy.
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### Components Involved

- **Electric Motor/Generator** – Converts kinetic energy into electrical energy.
  - **Battery Pack** – Stores the recovered electrical energy.
  - **Motor Controller (Inverter)** – Controls the flow of energy between the motor and battery.
  - **Battery Management System (BMS)** – Ensures safe charging of the battery.
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### Advantages of Regenerative Braking

- Improves the overall efficiency of the EV.
  - Increases the driving range by recovering energy.
  - Reduces energy consumption.
  - Minimizes wear on mechanical brake components.
  - Lowers maintenance costs.
  - Produces less heat during braking.
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### Limitations

- Cannot recover all the braking energy.
  - Less effective when the battery is fully charged.
  - Braking force is limited at very low speeds.
  - Mechanical brakes are still required for emergency and hard braking.
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### Comparison with Conventional Braking

| Regenerative Braking                         | Conventional Braking   |
|----------------------------------------------|------------------------|
| Recovers energy and stores it in the battery | Energy is lost as heat |

### Regenerative Braking

Improves vehicle efficiency

Reduces brake wear

Extends driving range

### Conventional Braking

Does not improve efficiency

Higher brake wear

No effect on driving range

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### Applications

Regenerative braking is commonly used in:

- Battery Electric Vehicles (BEVs)
- Hybrid Electric Vehicles (HEVs)
- Plug-in Hybrid Electric Vehicles (PHEVs)
- Electric buses and trains

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### Outcome

After completing this activity, students will be able to:

- Explain the working principle of regenerative braking.
- Identify the components involved in energy recovery.
- Compare regenerative braking with conventional braking.
- Understand how regenerative braking improves EV efficiency and driving range.

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### Conclusion

Regenerative braking is one of the key technologies that makes electric vehicles more energy-efficient. By converting kinetic energy into electrical energy during braking, it reduces energy loss, increases battery efficiency, extends vehicle range, and lowers maintenance costs. However, it works best alongside conventional braking systems to ensure safe and effective vehicle operation.

### Objective

To understand the factors affecting EV charging time and estimate charging duration.

### Introduction

Charging time depends on battery capacity, charger power, charging efficiency, battery state of charge, and temperature. A larger battery takes longer to charge, while a higher-power charger reduces charging time.

Formula:

Charging Time (hours) = Battery Capacity (kWh) ÷ Charger Power (kW)

Example: A 60 kWh battery charged using a 7 kW charger requires approximately 8.6 hours under ideal conditions. DC fast chargers significantly reduce charging time.

### Outcome

Students can estimate charging time for different battery capacities and charger ratings.

### Conclusion

Charging time is an important factor in EV adoption. Selecting the right charger improves convenience and charging efficiency.

#### **Activity 4: EV vs IC Engine**

##### **Objective**

To compare electric vehicles with conventional IC engine vehicles.

##### **Introduction**

Electric vehicles use electric motors powered by batteries, whereas IC engine vehicles use petrol or diesel engines. EVs offer higher efficiency, lower maintenance, zero tailpipe emissions, and quieter operation. IC engine vehicles have faster refueling and a more established fuel network but generate higher emissions and fuel costs.

##### **Outcome**

Students will compare the advantages and limitations of both technologies.

##### **Conclusion**

EVs are a cleaner and more efficient transportation solution, while ICE vehicles currently offer greater refueling convenience.

#### **Activity 5: Motors Used in EVs**

##### **Objective**

To study the different motors used in electric vehicles.

##### **Introduction**

The most common motors are Permanent Magnet Synchronous Motors (PMSM), Brushless DC (BLDC) motors, AC Induction Motors, and Switched Reluctance Motors (SRM). PMSMs provide high efficiency and torque density, BLDC motors are common in two-wheelers, induction motors are robust, and SRMs are simple and durable.

##### **Outcome**

Students will identify suitable motor types for various EV applications.

##### **Conclusion**

Motor selection directly affects the efficiency, performance, and cost of an EV.

#### **Activity 6: EV Infrastructure**

##### **Objective**

To understand the infrastructure required for reliable EV operation.

##### **Introduction**

An EV ecosystem requires a stable power supply, transformers, AC/DC chargers, parking space, communication systems, payment facilities, protection devices, and maintenance support. Renewable energy sources and battery storage systems can also be integrated to improve sustainability.

##### **Outcome**

Students will understand the essential infrastructure needed for EV charging.

##### **Conclusion**

Proper infrastructure ensures safe, reliable, and efficient operation of EV charging stations.

#### **Activity 7: Charging Station Location**

##### **Objective**

To identify suitable locations for highway charging stations.

##### **Introduction**

Charging stations should be installed every 50–100 km along highways, depending on EV range and traffic. Suitable locations include fuel stations, rest areas, restaurants, shopping centers, and places with adequate electrical infrastructure and security.



**Outcome**

Students will understand how charging station placement improves long-distance travel.

**Conclusion**

Strategic placement of charging stations reduces range anxiety and improves user convenience.

**Activity 8: Charging Station Rating****Objective**

To determine the required rating of an EV charging station.

**Introduction**

The station rating depends on the number of chargers, charger capacity, expected vehicle demand, diversity factor, and future expansion. The total connected load is calculated before selecting transformers, switchgear, and cables.

**Outcome**

Students will learn the basics of charging station sizing.

**Conclusion**

Correct station rating ensures reliable operation while avoiding unnecessary infrastructure costs.

**Activity 9: Installation Factors****Objective**

To study the factors affecting EV charging station installation.

**Introduction**

Important considerations include grid availability, transformer capacity, land availability, accessibility, safety standards, fire protection, drainage, communication network, government regulations, maintenance, and future scalability.

**Outcome**

Students will identify the major technical and economic factors influencing installation.

**Conclusion**

Proper planning ensures safe, economical, and future-ready charging stations.

**Activity 10: Charging Price****Objective**

To understand pricing strategies for EV charging.

**Introduction**

Charging prices should include electricity tariff, demand charges, maintenance, equipment cost, depreciation, taxes, and profit margin. Fast DC charging is generally priced higher than AC charging because of higher installation and operating costs.

**Outcome**

Students will understand how EV charging prices are determined.

**Conclusion**

A balanced pricing strategy ensures affordability for users while maintaining profitability for charging station operators.